

Problem

Often, when a list is sorted, the elements being sorted are just keys to other values. For example, if you are sorting files by their size, the sizes need to stay connected to their respective files. You cannot just take the size numbers and output them in order, you need to output all the required file information.

Submissions

The counting sort is used if you just need to sort a list of integers. Rather than using a comparison, you create an integer array whose index range covers the entire range of values in your array to sort. Each time a value occurs in the original array, you increment the counter at that index. At the end, run through your counting array, printing the value of each non-zero valued index that number of times.

Leaderboard

For example, consider an array **arr** = [1, 1, 3, 2, 1]. All of the values are in the range [0...3], so create an array of zeroes, **result** = [0, 0, 0, 0]. The results of each iteration follow:

Discussions

i	arr[i]	result
0	1	[0, 1, 0, 0]
1	1	[0, 2, 0, 0]
2	3	[0, 2, 0, 1]
3	2	[0, 2, 1, 1]
4	1	[0, 3, 1, 1]

Now we can print the sorted array: **sorted** = [1, 1, 1, 2, 3].

Challenge

Given an unsorted list of integers, use the counting sort method to sort the list and then print the sorted list.

Hint: You can use your previous code that counted the items to print out the actual values in order.

Function Description

Complete the countingSort function in the editor below. It should return the original array, sorted ascending, as an array of integers.

countingSort has the following parameter(s):

- arr: an array of integers

Input Format

The first line contains an integer **n**, the length of **arr**. The next line contains space-separated integers **arr[i]** where

$$0 \leq i < n.$$

Constraints

$$0 \leq n \leq 1000000$$

[Change Theme](#)

Language Python 3



```

15
16 def countingSort(arr):
17     # Write your code here
18     count = [0] * 100
19     for num in arr:
20         count[num] += 1
21     sorted_arr = []
22     for i in range(100):
23         sorted_arr.extend([i] * count[i])
24     return sorted_arr
25
26 if __name__ == '__main__':
27     fptr = open(os.environ['OUTPUT_PATH'], 'w')
28
29     n = int(input().strip())
30
31     arr = list(map(int, input().rstrip().split))
32
33     result = countingSort(arr)
34
35     fptr.write(' '.join(map(str, result)))
36     fptr.write('\n')
37
38     fptr.close()
39

```

Line: 17 Col: 27

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Test case 1

Test case 2

Test case 3

Test case 4

Compiler Message

Success